

Procedures for 5 Principals for Problem Solving

Process 1	Fact findings	Personnel resource	Cross Functional Team
		Material resource	No designation
INPUT			
The suppliers shall implement the following items to understand the current status of non-conformities and identify the root-causes of non-conformities in their products and quality management system. a) Observe and understand the facts on the basis of the Three Realities Principle (spot, thing, and situation) b) Review the facts by 5W2H and study factual information in detail - Who - What - When - Where - Why - How - How many / How much c) Analysis of the product based on information such as follows: - Information on nonconformity (information on symptom, situation including number of nonconformities and nonconformity rate and environment of problem occurrence) - Appearance of nonconforming products (visual check) - Comparison of dimensions with drawing specifications - Test result - Reproducibility analysis - Date of manufacture - Identification mark - Lot number d) Analysis of processes based on information such as follows: - Work shift, operators, etc. when problem occurred - Presence of abnormality in operation and / or equipment when problem occurred - Adequacy of listing in the Control plan and Operation standard, etc. - Quality records (daily report, inspection record, control chart, problem summary, etc.) - Nonconformity and repair history e) Classify information into the following categories based on the results of analysis, and identify the primary cause of the nonconformity: - Machine - Material - Manpower - Method - Others Note The suppliers begin investigation immediately to avoid impediments such as following factors: 1) Spot, thing, and situation change by the minute 2) Memories of personnel concerned begin to fade			
Process 2	Identifying causes (why-why analysis)	Personal resource	Cross Functional Team
		Material resource	No designation
INPUT			
The suppliers shall classify the identified causes of each symptom into the following categories and conduct a why-why analysis in aspects of both hardware side and software side. a) Occurrence b) Outflow c) Expansion In addition, why-why analysis must be conducted to identify the root causes of nonconformities, taking into consideration the followings. d) Repeatedly ask “why” as many times as necessary and analyze until the root cause is identified e) Conduct why-why analysis involving as many persons concerned as possible f) A cause for which all participants come to the same conclusion (such cause is most likely to be the root cause) g) Don't replace the cause with a person's problem The suppliers shall verify the validity of the why-why analysis results by considering the followings. h) Same or similar factors not repeatedly appeared i) Flow of why-why analysis not being reversed j) No conflict in the story from the occurrence of a problem to discovery of the root cause (no deviation from story)			

Note

How to proceed with Why-Why analysis: see **Figure 1**.

The benefits of performing why-why analysis include the followings.

- k) Level of problem can be clearly defined and communicated
- l) Relationship between multiple causes is easily understood
- m) Multiple causes can be identified
- n) Factors that can not be the cause can be identified along with the reasons
- o) Investigation process of causes is traceable
- p) Differences between individuals can be eliminated. Attention can be focused on the problem itself, not the history of the problem
- q) Attention can be focused on causes, not symptoms
- r) Support systems for problem solving can be established by gaining consent from related sections and persons, etc.

Process 3	Appropriate measures	Personal resource	Cross Functional Team
		Material resource	No designation
INPUT			
The suppliers shall implement countermeasures to the identified root causes, taking into account the followings.			
a) Examine measures, whenever possible, in aspects of both hardware side and software side			
b) Measures shall be appropriate not only for the process or operation concerned but also for the organization in totality (systematic corrective actions)			
c) Implement measures focusing more on the elimination of cause, rather than prevention of outflows			
Measures shall be employed on a permanent basis (elimination of causes / corrective actions). If judged necessary based on the degree of severity, urgency, and negative effects, etc., take interim measures (containment of nonconforming products / containment actions).			
Timing to implement permanent measures shall be defined in advance when interim measures are taken.			
Process 4	Verification of effectiveness of measures	Personal resource	Cross Functional Team
		Material resource	No designation
INPUT			
The suppliers shall evaluate the effectiveness of the applied corrective actions, taking into account the followings.			
a) Verify effectiveness of measures with reasonable number of products or time period, and confirm no recurrence of problems from the same cause			
b) Verify both products and processes			
(e.g.) Product: disassemble completed products for verification, if necessary.			
Process: verify the process in which problems were found.			
c) Effectiveness shall be verified per shift, operator, etc., depending on the cause of problems or the nature of measures			
d) Verify that the results of measures will not have any adverse effect on other products, processes, operations, etc.			
Process 5	Feedback to upstream sources	Personal resource	Cross Functional Team
		Material resource	No designation
INPUT			
The suppliers shall update the following documents based on corrective actions, as needed, but are not limited to the following:			
a) Results of risk analysis (FTA、DFMEA、PFMEA)			
b) Process flow chart			
c) Control plan			
d) Operation standard			
Also, information including the followings shall be shared with related organizations and sections to prevent nonconformity from occurring in similar operations, processes or systems.			
e) Sections / areas in which similar operations, processes, or systems are used			
f) Section found the nonconformity			
g) Section responsible for designing processes			
h) Section responsible for designing products			
i) Section responsible for controlling quality system documents			

Figure 1 Why-why Analysis

Start from the problem symptom. Sort out and systematize relevant factors until the root cause is revealed.

